

PLAN OF WORK

Driverless Vehicle Edge-Case Intelligence Platform

Machine Learning and Full-Stack Delivery Plan

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Scope: Dashcam risk-event analysis, incident knowledge base, and web platform delivery
Workstreams: Machine learning pipeline; full-stack platform and data infrastructure

Objective: Build a structured edge-case intelligence platform that can identify traffic-risk events from ego-centric dashcam videos, extract the most relevant incident clips, classify the core outcome, generate semantic tags and summaries, and present the results in a searchable and annotatable web system.

Scope at a Glance

Area	Plan
Machine learning	Process ego-centric dashcam video, localise peak-risk moments, extract event-centred clips, classify outcomes, and generate tags and summaries.
Full-stack platform	Provide upload, storage, asynchronous processing, searchable incident records, map visualisation, timeline review, and human annotation workflows.
Target clip set	Approximately 3,000-3,500 fixed-length clips, each around 10 seconds, with the event time or peak-risk timestamp near the centre.
Primary technologies	BADAS, Qwen3-VL, FastAPI, AWS S3, Celery/Redis, PostgreSQL, React, TypeScript, Mapbox, and Chart.js.

1. Machine Learning Workstream

The machine learning workstream takes ego-centric dashcam videos as input and produces structured traffic-risk analysis outputs. The workstream is organised around four core functions.

- Risk event detection.** Detect whether a video contains an ego-relevant traffic-risk event.
- Key clip extraction.** Identify the event timestamp or peak-risk moment, then extract a fixed-length event-centred clip.

3. **Core outcome classification.** Classify the extracted clip as safe, near_miss, or collision.
4. **Semantic tagging and summary generation.** Assign structured semantic tags and generate a short incident summary for the extracted event.

Stage 1: Dataset Standardisation and Clip-Level Labelling

The first stage standardises the available datasets into a unified clip-level format. Nexar and SAVeD will be used as the main datasets. NIDB remains optional and will only be used if additional data is required.

Dataset	Role	Labelling implication
Nexar	Main target-domain dataset.	Provides event timing for positive cases, which can supervise event-centre localisation. Positive samples require manual relabelling because collision and near-miss events are merged into one positive class.
SAVeD	External hard-sample dataset.	Provides crash and near-miss examples with richer semantic annotations, improving recognition of difficult edge cases and supporting downstream semantic tags.
NIDB	Optional supplementary dataset.	Used only if the target clip count or class balance requires additional data.

1. Unified core label schema: safe, near_miss, collision.
2. Target dataset size: approximately 3,000-3,500 fixed-length event-centred clips.
3. Clip construction: around 10 seconds per clip, with the event time or peak-risk timestamp near the centre.
4. Event centre source: Nexar event timing for Nexar; BADAS peak-risk time for SAVeD and other datasets without reliable timestamps.

Stage 2: BADAS-Based Risk Detection, Clip Extraction, and Classification

The first ML model will be BADAS-based. It provides the temporal backbone of the ML pipeline because it is designed around risk prediction in driving videos and can generate a temporal risk signal over a clip.

Model element	Specification
Responsibilities	Risk timeline prediction; key clip extraction; safe / near_miss / collision classification.
Input clip	10-second clips sampled at 8 fps, producing approximately 80 frames per clip.
Internal temporal window	Overlapping 16-frame windows, corresponding to approximately 2 seconds per window.
Outputs	Predicted core classification label, risk timeline, and extracted 10-second key clip for semantic understanding.

This stage identifies not only whether risk exists, but also when the key risk moment occurs. That timing signal supports event detection, localisation, and event-centred clip extraction before semantic interpretation begins.

Stage 3: Qwen3-VL-Based Semantic Tagging and Summary Generation

The second ML model will be a Qwen3-VL-based video tag classifier. It receives the BADAS-extracted 10-second clip and performs richer semantic interpretation rather than frame-level temporal risk prediction.

1. Semantic understanding of the extracted incident clip.
2. Structured tag prediction across actor type, scenario type, road type, weather, avoidance behaviour, collision geometry, and near-miss type.
3. Short structured incident summary generation.
4. Frame sampling tests in the expected range of 2-5 fps to balance semantic detail against computational cost.

ML architecture principle: BADAS handles temporal risk detection, localisation, clip extraction, and core classification. Qwen3-VL handles semantic interpretation and structured description. This separation makes evaluation, debugging, and later model improvement more manageable.

2. Full-Stack Workstream

The full-stack workstream delivers the web application and supporting data infrastructure. Its purpose is to ingest uploaded videos, trigger ML processing, store analysis outputs, and present incidents to users in a structured and searchable format.

Backend

The backend will be developed using Python, FastAPI, AWS S3, and Celery/Redis. Upload and inference are separated so that long-running ML jobs do not block the user interface.

1. **Validate and store upload.** The backend validates the video file and stores it in S3.
2. **Create incident record.** A database record is created to track the uploaded video and its processing state.
3. **Dispatch background job.** Celery/Redis starts asynchronous ML processing without blocking the frontend.
4. **Write ML outputs.** The worker writes the core classification, risk timeline, semantic tags, and generated summary back to the incident record.

API endpoints will allow the frontend to filter incidents by classification, semantic tag, date range, location, and other relevant metadata. The backend will also support human annotation overrides, recording the old value, new value, and reviewer identity for every correction.

Database

The database will use PostgreSQL. It will store original uploaded metadata, ML-generated outputs, and human annotation changes so the platform can operate as a structured incident knowledge base.

Table	Purpose
incidents	Primary incident records, uploaded video metadata, location, status, and storage references.
labels	Core model and human-reviewed classification values: safe, near_miss, or collision.
tags	Structured semantic tags produced by Qwen3-VL and adjusted by reviewers when needed.
risk_timeline	Temporal risk signals produced by BADAS for incident review and chart visualisation.
summaries	Generated incident summaries linked to the analysed clip.
label_changes	Audit history for manual overrides, including old value, new value, reviewer identity, and timestamp.

Frontend

The frontend will be developed using React, TypeScript, Mapbox, and Chart.js. It will provide the operational interface for upload, search, review, mapping, timeline inspection, and annotation.

1. Drag-and-drop video upload with optional location pin, narrative description, and context tags.
2. Processing status display for waiting, processing, and completed incidents.
3. Interactive incident map with classification-coloured pins and filters for classification, tags, dates, and location.
4. Incident detail view showing the related video, risk timeline, semantic tags, and generated summary.
5. Queryable and sortable incident database view for structured search and inspection.
6. Annotation interface for overriding model-generated classifications and tags, with corrections written back to the database.

3. Justification of the Proposed Approach

The proposed approach separates temporal risk detection from semantic interpretation. This keeps the ML system modular and aligns each model with the task it is best suited to perform.

1. BADAS is used for detecting risk events, generating a risk timeline, extracting key clips, and classifying the core outcome because these tasks depend on locating the riskiest moment within the video.
2. Qwen3-VL is used for semantic tag classification and structured summary generation because these tasks require higher-level visual and contextual understanding, including actor types, road layouts, weather conditions, avoidance behaviour, and incident geometry.

The full-stack architecture supports the practical workflow of the platform. FastAPI provides a lightweight API layer; AWS S3 provides scalable video storage; Celery/Redis enables asynchronous ML jobs; PostgreSQL provides structured incident storage; React and TypeScript support a maintainable frontend; and Mapbox and Chart.js support map-based visualisation and risk timeline display.

End-to-End Workflow

1. **Upload.** A user uploads a driving video and optional context metadata.
2. **Store and queue.** The backend stores the video, creates the incident record, and queues asynchronous ML processing.
3. **Detect and extract.** BADAS predicts risk, localises the key moment, extracts the event-centred clip, and classifies the core outcome.
4. **Interpret.** Qwen3-VL assigns semantic tags and generates a concise incident summary.
5. **Review and manage.** Users search, filter, map, inspect, and annotate incidents through the web platform.

Resulting Platform Outputs

Output	Description
Analysed incident record	A structured database record linked to the original upload and extracted key clip.
Risk timeline	Temporal risk signal suitable for chart-based review.
Semantic tags	Structured incident descriptors for search, filtering, and dataset analysis.
Generated summary	Short natural-language description of the incident.
Annotation audit trail	Recorded human corrections that can support quality control and future model improvement.